$$P(y) = \int P(y|x)\, P(x)\, \mathrm{d}x$$

Introduction to LaTeX

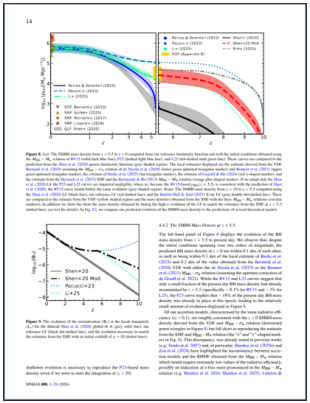
$$x_{n+1} = \lambda x_n (1 - x_n)$$

$$\lim_{x \to \infty} \frac{1}{1 + x^{-1}}$$

$$\sum_{n=1}^{N} n(n-1)$$

Session 3: Tables, Bibliography and Mathematics

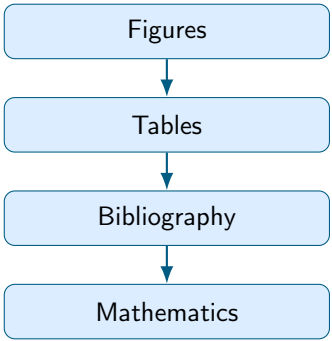
$$\int_b^a f(x)\, \mathrm{d}x$$



- Organising large documents: `\input`, `\include`, folders, custom packages.
- Using packages: `graphicx`, `geometry`, `hyperref`, and friends.
- Figures: `figure`, `\includegraphics`, captions, labels, and cross-references.
- Float behaviour: `h`, `t`, `b`, `p`, `!`, and `H`.

Outline

- What carries over from figures?
- Tables: `tabular`, `table`, `booktabs`
- Bibliography: `.bib`, `biblatex`, citation commands
- Mathematics: inline, display, numbered equations, `amsmath`
- Short exercise after each topic
- Homework



Tables, Bibliography and Mathematics

The same pattern as figures: tables and equations



Object	Label prefix	Typical text reference
Figure	`fig:`	Figure `\ref{fig:...}`
Table	`tab:`	Table `\ref{tab:...}`
Equation	`eq:`	Equation `\ref{eq:...}`

1 Tables

Why tables?

Use a table when

- exact values matter;
- the reader needs to compare rows or columns;
- results are too small for a plot;
- you need a compact summary.

Compact summary

Model	Acc.	Time
A	0.81	12.4
B	0.88	15.2

Source: Overleaf tables guide (Overleaf 2026d).

Tables: source code → output

```
\begin{tabular}{lcc}
  \toprule
  Model & Accuracy & Time (s) \\
  \midrule
  Baseline & 0.81 & 12.4 \\
  Proposed & 0.88 & 15.2 \\
  \bottomrule
\end{tabular}
```

Model	Accuracy	Time (s)
Baseline	0.81	12.4
Proposed	0.88	15.2

Key ideas

- Column alignment is defined in `{lcc}`.
- Cells are separated by `&`.
- Rows are separated by `\\`.
- Horizontal rules are created with `\toprule`, `\midrule`, and `\bottomrule`.

Source: Adapted from Overleaf (Overleaf 2026d); `booktabs` (CTAN 2026b).

The two layers: `tabular` vs `table`

Inside: content and alignment

```
\begin{tabular}{lcc}
  ... rows and columns ...
\end{tabular}
```

- `l` (left), `c` (centre), `r` (right): column alignment.

Outside: caption, label, float

```
\begin{table}[htbp]
  \centering
  ... tabular ...
  \caption{Short description.}
  \label{tab:results}
\end{table}
```

- `table` behaves like `figure`.
- Put `\label` after `\caption`.
- The `\caption` may appear above or below the `tabular` environment.

Source: Lib4RI, *Figures, Tables and Formulae* (Hösli 2025).

Building a table step by step

```
\begin{table}[htbp]          % 1. float
\centering                   % 2. centre
\begin{tabular}{lcc}         % 3. columns
\toprule                     % 4. booktabs rules
Model & Accuracy & Time (s) \\
\midrule
Baseline & 0.81 & 12.4 \\
Proposed & 0.88 & 15.2 \\
\bottomrule
\end{tabular}
\caption{Example results table using
        \texttt{booktabs}.} % 5. caption
\label{tab:results}          % 6. label
\end{table}
```

1. Start the table environment.
2. Centre the table.
3. Declare columns.
4. Use `booktabs` rules.
5. Add a meaningful caption.
6. Add a meaningful label.

Avoid

Model	Acc.	Time
A	.81	12.4
B	.88	15.2

Prefer

Model	Accuracy	Time (s)
Baseline	0.81	12.4
Proposed	0.88	15.2

Rule of thumb

Use fewer rules, more spacing, clear headings, and units in the column header.

Source: booktabs guidance (CTAN 2026b).

Referencing tables

```
\begin{table}[htbp]
  \centering
  ... tabular ...
  \caption{Example results table using
        \texttt{booktabs}.}
  \label{tab:results}
\end{table}

Table~\ref{tab:results} uses the
\texttt{booktabs} package.
```

Same logic as figures

- Give the table a `\caption` and a `\label`.
- Refer to the label in the text.
- Compile twice if references show ??.

Table template: code

```latex
Table~\ref{tab:results} uses the \texttt{booktabs} package. A good table
avoids unnecessary vertical lines and places units in the column headings.

\begin{table}[htbp]          % 1. float
\centering                   % 2. centre
\begin{tabular}{lcc}         % 3. columns
\toprule                     % 4. booktabs rules
Model & Accuracy & Time (s) \\
\midrule
Baseline & 0.81 & 12.4 \\
Proposed & 0.88 & 15.2 \\
\bottomrule
\end{tabular}
\caption{Example results table using
        \texttt{booktabs}.} % 5. caption
\label{tab:results}          % 6. label
\end{table}
```

Table template: output

Table 1 uses the `booktabs` package. A good table avoids unnecessary vertical lines and places units in the column headings.

Model	Accuracy	Time (s)
Baseline	0.81	12.4
Proposed	0.88	15.2

Table 1: Example results table using `booktabs`.

Exercise: build and reference a table

In your Session 2 report

1. Add `\usepackage{booktabs}` to the preamble.
2. Create one table using `table` and `tabular`.
3. Use `\toprule`, `\midrule`, and `\bottomrule`.
4. Add a meaningful `\caption` and `\label`.
5. Refer to it in the text using `Table~\ref{tab:...}`.

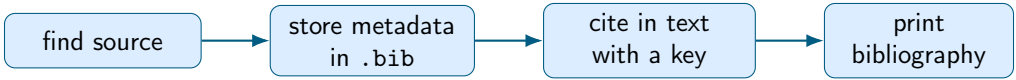
Checkpoint

Recompile until the table reference shows the correct number instead of ??.

2

Bibliography

The citation workflow



- The `.bib` file is a database, not a manually typed reference list.
- The citation key links the source metadata to your writing.
- The bibliography style handles formatting.

Source: Overleaf bibliography management with `biblatex` (Overleaf 2026b).

BibTeX/natbib or biblatex?

Approach	Strength	When to use
BibTeX + natbib	Very widely supported; familiar author–year and numeric citations	Many journal and older document templates
biblatex + Biber	Modern, flexible, and easier to customise	New projects, reports, and theses unless a template says otherwise

Course recommendation

Use biblatex unless a journal or thesis template requires something else.

Minimal biblatex setup

```
% main.tex preamble
\usepackage[backend=biber,style=authoryear,
sorting=nyt,dashed=false]{biblatex}
\addbibresource{references.bib}
```

```
% main.tex body
Recent work discusses this topic
\parencite{smith2024}.

\printbibliography
```

```
% references.bib
@article{smith2024,
  author       = {Smith, Jane},
  title        = {A Useful Paper},
  journaltitle = {Journal Name},
  year         = {2024}
}
```

Key ideas

- `backend=biber` tells `biblatex` to use Biber.
- `style=authoryear` produces author–year citations.
- `sorting=nyt` sorts by name, year, and title.
- `dashed=false` repeats an author's name in each entry.
- `\addbibresource` loads the `.bib` file.
- `\printbibliography` prints the reference list.

Source: Overleaf `biblatex` guide (Overleaf 2026b).

Citation commands students will use often

Command	Output idea	Use case
`\textcite{key}`	Author (year)	Source is part of the sentence
`\parencite{key}`	(Author, year)	Source supports the claim
`\cite{key}`	style-dependent	General citation command
`\autocite{key}`	style-dependent	Adapts across styles

Example

`\textcite{smith2024}` argues that...
This method is widely used `\parencite{smith2024}`.

Bibliography template: code

main.tex × references.bib

Code Visual

```latex
\documentclass{article}
\usepackage[colorlinks=true,linkcolor=blue,citecolor=blue,urlcolor=blue]
{hyperref}

\usepackage[backend=biber,style=authoryear,sorting=nyt,dashed=false]
{biblatex}
\addbibresource{references.bib}

\title{Simple Bibliography Template}
\author{Your Name}
\date{\today}

\begin{document}

\maketitle

\section{Introduction}

This is to show how to cite a paper: \parencite{smith2024}.\\
This is to show how to cite a \textit{wrong} paper: \parencite{smith2025}.

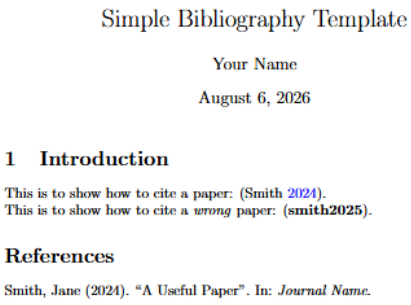
\printbibliography

\end{document}
```

main.tex references.bib ×

```bibtex
@article{smith2024,
  author = {Smith, Jane},
  title  = {A Useful Paper},
  journaltitle = {Journal Name},
  year   = {2024}
}
```

Bibliography template: output



Simple Bibliography Template

Your Name

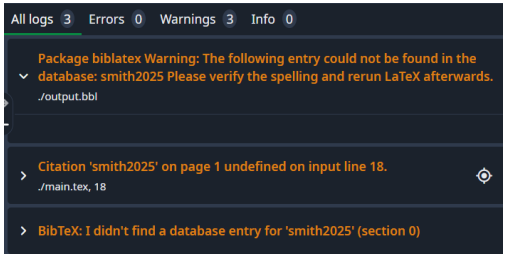
August 6, 2026

1 Introduction

This is to show how to cite a paper: (Smith 2024).
This is to show how to cite a *wrong* paper: **(smith2025)**.

References

Smith, Jane (2024). "A Useful Paper". In: *Journal Name*.



Warning!

You will receive a warning if you cite a reference that is not in the `.bib` file.

Choose the entry type that matches your source

Entry type	Use for	Typical fields
`@article`	Journal article	author, title, journaltitle, year
`@book`	Book	author, title, publisher, year
`@incollection`	Chapter in an edited book	author, title, booktitle, editor
`@inproceedings`	Conference paper	author, title, booktitle, year
`@thesis`	PhD/MPhil/MSc thesis	author, title, institution, year
`@report`	Research/technical report	author, title, institution, year
`@online`	Website or online resource	author, title, url, urldate

Key idea

The word after `@` tells `biblatex` what kind of source it is. Different entry types use different metadata fields.

Source: Overleaf `biblatex` guide (Overleaf 2026b).

Exercise: add citations and a bibliography

In your report

1. Add two bibliography entries of different types to `references.bib`.
2. Add the `biblatex` setup and `\addbibresource{references.bib}`.
3. Cite the source in the text using `\textcite` or `\parencite`.
4. Add another citation somewhere in the report.
5. Print the reference list using `\printbibliography`.

Checkpoint

The citation and bibliography entry should both appear correctly after recompiling.

3 Mathematics

Inline and display mathematics

```
Inline: the loss is $L(\theta)$.

Display, unnumbered:
\[
  L(\theta)=\sum_{i=1}^{n}
  (y_i-f_\theta(x_i))^2.
\]

Display, numbered:
\begin{equation}
  L(\theta)=\sum_{i=1}^{n}
  (y_i-f_\theta(x_i))^2.
\end{equation}
```

Inline: the loss is $L(\theta)$.

Display, unnumbered:

$$L(\theta) = \sum_{i=1}^{n} (y_i - f_\theta(x_i))^2.$$

Display, numbered:

$$L(\theta) = \sum_{i=1}^{n} (y_i - f_\theta(x_i))^2. \quad (1)$$

Source: Overleaf mathematical expressions (Overleaf 2026c); Lib4RI, *Figures, Tables and Formulae* (Hösli 2025).

Use amsmath for structured equations & amssymb for extra symbols

```
% main.tex preamble
\usepackage{amsmath,amssymb}
```

```
% main.tex body
\begin{align}
  y_i &= \beta_0+\beta_1 x_i+\epsilon_i, \\
  \hat\beta &= (X^T X)^{-1}X^T y.
\end{align}

\[
f(x)=
\begin{cases}
  1,   & x=0, \\
  x^2, & x\in\mathbb{R}\setminus\{0\}.
\end{cases}
\]
```

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \tag{2}$$

$$\hat{\beta} = (X^T X)^{-1} X^T y. \tag{3}$$

$$f(x) = \begin{cases} 1, & x = 0, \\ x^2, & x \in \mathbb{R} \setminus \{0\}. \end{cases}$$

Tip

Use `align*` instead of `align` if equation numbers are not needed.

Source: amsmath documentation (CTAN 2026a).

What do amsmath and amssymb provide?

amsmath: structured mathematics

- Provides environments for writing and arranging equations.
- `align` aligns several equations at a chosen point, usually the =.
- `cases` typesets piecewise definitions with a left brace.

amssymb: additional mathematical symbols

- Provides additional mathematical symbols and fonts.
- For example:

$$x \in \mathbb{R}, \qquad \varnothing, \qquad A \subseteq B.$$

- It is commonly loaded together with `amsmath`.

Source: AMS mathematical packages; `amsmath` documentation (CTAN 2026a).

A few common examples

Type	Output	LaTeX input
Greek letters	α, β, γ	`\alpha`, `\beta`, `\gamma`
Relations	$\leq$, $\geq$, $\neq$	`\leq`, `\geq`, `\neq`
Sets	$\in$, $\subseteq$, $\cup$	`\in`, `\subseteq`, `\cup`
Number sets	$\mathbb{R}$, $\mathbb{N}$	`\mathbb{R}`, `\mathbb{N}`
Operators	$\times$, $\pm$, $\sum$, $\int$	`\times`, `\pm`, `\sum`, `\int`
Arrows	$\rightarrow$, $\leftrightarrow$	`\rightarrow`, `\leftrightarrow`

Useful lookup pages

- Overleaf: List of Greek letters and math symbols

Source: Overleaf list of Greek letters and math symbols (Overleaf 2026a).

```
\begin{equation}
\label{eq:loss}
  L(\theta)=\sum_i (y_i-f_\theta(x_i))^2
\end{equation}

Equation~\ref{eq:loss} defines the loss.
```

$$L(\theta) = \sum_i (y_i - f_\theta(x_i))^2 \tag{4}$$

Equation 4 defines the loss.

Remember

- Put `\label` inside the numbered equation environment.
- Use a meaningful label, e.g. `eq:loss`.
- Use `\ref` to refer to the equation number.

The equation is part of the sentence

- Introduce the equation before displaying it.
- Define symbols before using them, especially in interdisciplinary writing.
- Number only equations that you will reference later.
- Write text between equations to explain the transition.

Good habit

Write the sentence first, then insert the equation where it supports the sentence.

Writing mathematics template: code

```
35 Inline mathematics appears inside a sentence, such as $x \in \R^d$ or
36 $L(\theta)$. Display mathematics is centred and easier to read:
37
38 \[
39   L(\theta) = \sum_{i=1}^{n} \left(y_i - f_\theta(x_i)\right)^2.
40 \]
41
42 Numbered equations should be used when you need to refer to them later. For
43 example, Equation~\ref{eq:linear-model} defines a simple linear model:
44
45 \begin{equation}\label{eq:linear-model}
46   y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.
47 \end{equation}
48
49 For aligned equations, use the \texttt{align} environment from
   \texttt{amsmath}:
50
51 \begin{align}
52   \hat{\beta} &= (X^\top X)^{-1}X^\top y, \\
53   \norm{X\hat{\beta}-y}^2 &= \sum_{i=1}^{n}\left(x_i^\top\hat{\beta}-
     y_i\right)^2.
54 \end{align}
```

2.3 Inline, display, and numbered mathematics

Inline mathematics appears inside a sentence, such as $x \in \mathbb{R}^d$ or $L(\theta)$. Display mathematics is centred and easier to read:

$$L(\theta) = \sum_{i=1}^{n} (y_i - f_\theta(x_i))^2 .$$

Numbered equations should be used when you need to refer to them later. For example, Equation 1 defines a simple linear model:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i. \tag{1}$$

For aligned equations, use the `align` environment from `amsmath`:

$$\hat{\beta} = (X^\top X)^{-1} X^\top y, \tag{2}$$

$$\left\| X\hat{\beta} - y \right\|^2 = \sum_{i=1}^{n} \left(x_i^\top \hat{\beta} - y_i \right)^2 . \tag{3}$$

Exercise: write and reference mathematics

In your report

1. Add at least one inline mathematical expression.
2. Add one unnumbered display equation.
3. Add one numbered equation using the `equation` environment.
4. Give the numbered equation a meaningful `\label`.
5. Refer to it in the text using `Equation~\ref{eq:...}`.

Optional challenge

Use the `align` environment to typeset two related equations aligned at the = sign.

4

Homework

Complete and polish the report you worked on during today's exercises. It should contain:

- figures from Session 2;
- tables with captions and cross-references;
- a reference database and bibliography;
- citations in the main text;
- inline and display mathematics;
- at least one numbered and referenced equation.

For next session

Bring the report content. We will convert it into a short Beamer presentation.



Extra Slides

Quick troubleshooting checklist

Symptom	First thing to check
Misplaced alignment tab	Too many or too few `&`; check column specification.
Citation undefined	Wrong citation key, missing `.bib` entry, or Biber not run.
Cross-reference shows ??	Compile again; check the `\label` and `\ref` keys match.
Missing $ inserted	Math or underscore outside math mode.
Table too wide	Reduce columns, abbreviate headings, or use smaller table text.

Alternative workflow

Do not load `natbib` together with `biblatex`. For this workflow, use `\usepackage{natbib}` instead.

Command	Output idea
`\citet{key}`	Author (year)
`\citep{key}`	(Author, year)
`\citealt{key}`	Author year
`\citealp{key}`	Author, year

Example

`\citet{smith2024}` argues that...

This method is widely used `\citep{smith2024}`.

Source: `natbib` documentation (Daly 2026).

Paper Aliases

`natbib` lets you give a reference a short alias, for example *Paper I*, and use that alias later in the text.

Command	Use
`\defcitealias{key}{alias}`	Define an alias for a reference
`\citet[][hereafter \citetalias{key}]{key}`	Introduce the alias at the first citation
`\citetalias{key}`	Print the alias in text
`\citepalias{key}`	Print the alias in parentheses

Key idea

Define the alias once, introduce it at the first citation, then use the short alias later.

Source: `natbib` documentation (Daly 2026).

Citation aliases: example

```
% preamble
\defcitealias{smith2024}{Paper~I}

% main text
The first paper in this series, \citet[][hereafter \citetalias{smith2024}]{smith2024},
     demonstrated that...\\

Here, we build upon \citetalias{smith2024}, extending the framework to ...
```

Output

The first paper in this series, Smith et al. (2024, hereafter Paper I), demonstrated that...

Here, we build upon Paper I, extending the framework to...

Source: `natbib` documentation (Daly 2026).

CTAN (2026a). *amsmath – AMS Mathematical Facilities for LaTeX*. URL: https://ctan.org/pkg/amsmath (visited on 09/15/2026).

CTAN (2026b). *booktabs – Publication Quality Tables in LaTeX*. URL: https://ctan.org/pkg/booktabs (visited on 09/15/2026).

Daly, Patrick W. (2026). *Natural Sciences Citations and References: The natbib Package*. URL: https://mirrors.ctan.org/macros/latex/contrib/natbib/natbib.pdf (visited on 09/15/2026).

Hösli, Frank (2025). *Getting Started with LaTeX: Figures, Tables and Formulae*. URL: https://www.lib4ri.ch/sites/default/files/media/documents/2025_04_LaTeX_2_figures_tables_math.pdf (visited on 09/15/2026).

Overleaf (2026a). *List of Greek Letters and Math Symbols*. URL: https://www.overleaf.com/learn/latex/List_of_Greek_letters_and_math_symbols (visited on 09/15/2026).

Overleaf (2026b). *Bibliography Management with BibLaTeX*. URL: https://www.overleaf.com/learn/latex/Bibliography_management_with_biblatex (visited on 09/15/2026).

Overleaf (2026c). *Mathematical Expressions*. URL: https://www.overleaf.com/learn/latex/Mathematical_expressions (visited on 09/15/2026).

Overleaf (2026d). *Tables*. URL: https://www.overleaf.com/learn/latex/Tables (visited on 09/15/2026).

Thank You